

Forager

Immersive Mycology Experiences and Time-lapse Photogrammetry

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ABSTRACT

Forager is a multisensory virtual reality experience that immerses participants in the captivating world of fungi, delivering a profound connection with nature from various perspectives. Engaging sight, sound, touch, and scent, participants witness the complete life cycle of mushrooms, from spores to mycelium, fruiting body, and the inevitable decay. This transformative journey cultivates an appreciation for the interconnectedness of lifeforms and the wonders of the natural world on its own unique time scale.

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1 INTRODUCTION

Time-lapse photography and photogrammetry have greatly enhanced our ability to observe and understand the world around us. However, these two techniques have never been combined before. Forager developed a software and hardware solution to merge these two techniques into time-lapse photogrammetry and used this solution to capture a variety of mushrooms growing and decaying over time. We opted to present these sequences as a narrative virtual reality experience, as this medium allows the viewer to witness an elevated sense of immersion. Playing back the time-lapse photogrammetry sequences in full detail and fast frame rate required the use of a custom data format that was developed for real-time processing.

By leveraging accurate visualization techniques, Forager aims to uncover the untapped potential of mycology while showcasing their natural allure, ultimately fostering a deeper understanding and appreciation for the wonders of fungi.

This paper focuses on elucidating the procedural aspects pertaining to the tools and pipelines involved in generating time-lapse photogrammetry, as well as creating the immersive elements of the Forager experience.

2 MOTIVATION

The project was inspired by Porter's relocation to Maine during the pandemic, where he observed a thriving mushroom culture and the potential for immersive technology to connect people with nature. Growing his own Blue Oyster mushrooms further fueled his motivation to capture the mushrooms' rapid growth and voluminous beauty in 3D through a combination of photography techniques.

Growing mushrooms are also the perfect subject when considering growth rate, size, complexity, aesthetics, and availability. Our technology allows participants to observe kingdom Fungi in a way that was previously not possible with the naked eye.

3 TIME-LAPSE PHOTOGRAMMETRY

Automation was a key goal in our development plan, as it was necessary to trigger captures in a consistent fashion and on a regular time interval. Executing photogrammetry manually can be an arduous process, and having consistent captures allowed us to also automate the reconstruction and meshing steps of the pipeline.

3.1 Capture

A handheld photogrammetry test using a grocery store mushroom yielded impressive results, prompting the development of an automated image capture system. Using a Raspberry Pi microcontroller, a motorized turntable, and multiple cameras arranged in a column, we achieved consistent and efficient image captures of the subject from all angles. The software running on the microcontroller was written in a modular way to accept any number of cameras and angles per capture.

3.2 Rig

Through iterative prototype development, we designed a rig (see Figure 1) that has a good balance of capture angles and minimizing occlusion. Because we are capturing an animation over time, all frames must share a 3D space, with a constant position, orientation, and scale of the subject. We achieve this using image markers on the rig, placed at different positions and angles. These are detected by the photogrammetry software RealityCapture and help it achieve an accurate spatial understanding.

3.3 Reconstruction

After trying different photogrammetry solutions, we opted for RealityCapture, which offered superior image quality and a robust command line interface. To increase efficiency and handle a high number of captures, we developed a tool called "MycoMachine", a GUI for generating command line instructions for RealityCapture. After initialization, MycoMachine operates on a watch folder, eliminating the need for manual input, and processing the captures off the Raspberry Pi into RealityCapture automatically.

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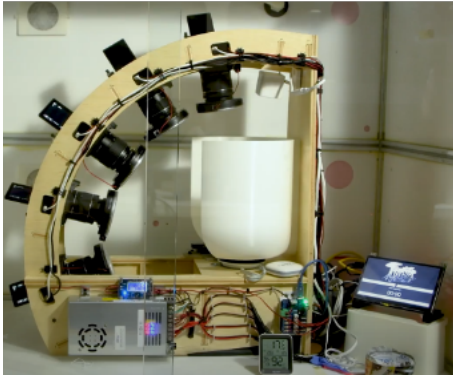


Figure 1: Camera Rig

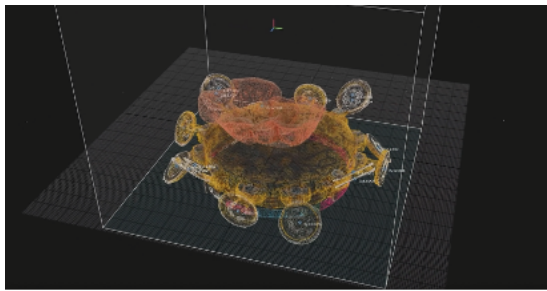


Figure 2: Model in RealityCapture

3.4 Model Cleanup

After reconstruction, the next step is to clean up the model, removing extraneous elements from the output geometry such as the rig and registration markers. Even under controlled conditions, mushrooms have irregular and unpredictable growth, so we could not simply use rectangular or spherical bounds to determine which parts of the mesh to keep and which to discard. We used Houdini to generate a custom boundary volume for each mushroom sequence, and automated the process of cleaning up the frames using custom Python nodes directly in Houdini.

Discarding the extra geometry made room for extra space in the model texture, which could be used to increase the resolution of the actual mushroom surface texture. We take advantage of this by unwrapping the UVs of the clean model in Houdini, then reprojecting the textures back in RealityCapture, enabling access to the original high-resolution images (see Figure 2).

3.5 Playback Format

We wanted to have the least restrictions for sequence playback, including playback direction and speed, frame rate, mesh density, and texture resolution. We ran tests with common formats like Alembic streams, OBJ sequences, FBX sequences but none offered the flexibility we desired. We achieved the best results by converting each mesh to a dense point cloud and encoding the data in EXR images. We use one EXR image per model frame, and stream the images from disk at runtime, keeping a preloaded cache in memory. Each image can hold up to 1024x1024 data points, including position,

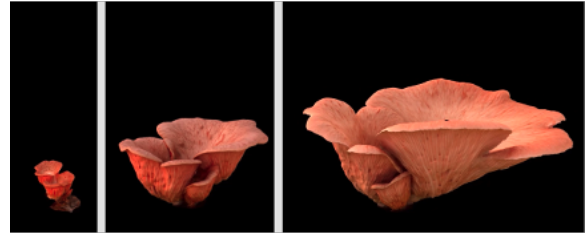


Figure 3: Volumetric Pink Oyster



Figure 4: Photogrammetry mushroom in Unreal Engine

color, and normal data, which allows us to light the point cloud in the renderer. We encode the EXR files in Houdini using a batch script and decode the frame in Unreal Engine's Niagara system to reconstruct the point cloud in real-time (see Figure 3).

4 IMMERSIVE EXPERIENCE

Using the Unreal Engine 5 game engine and Dell Precision workstations, we transformed the captured time-lapse photogrammetry sequences into an immersive story (see Figure 4) where users can experience the complete life-cycle of a mushroom organism. The journey unfolds in four phases: the release of the spore, the mycelial growth in the soil, the fruiting body and the decay caused by the passing of time. The final scene hints at the pressing environmental issues caused by urbanization and loss of old-growth forests. Through this narrative, we aim to uncover the often-overlooked impact mushrooms have on shaping our environment, as well as the indirect yet impactful role human activities play in the premature decay and decline of essential fungal colonies.

4.1 Multisensory Elements

Along with the standard virtual reality elements of stereoscopic visuals and spatial audio, Forager employs technology which engages other senses to help elevate the audience's sense of immersion. Wind is a common method for spore dispersal, so we used custom Arduino-controlled power switches to control fans which emulate a forest breeze. Gusts are carefully timed to sync up with appropriate visuals inside the headset. In order for the participants to feel a sense of the weightlessness of a spore, we have them laying down on Moon Pod beanbags. These are equipped with haptic transducers from Subpac. The low frequency rumbles help the audience feel



Figure 5: OW Scent Delivery Device



Figure 6: Mycelial bricks structure

thunderstorms, mycelial growth and the meditative soundscape, created using the Mach1 spatial audio technology by Q Department. Forager also utilizes scent to help participants further embody the life cycle of a mushroom. Using OW Smell Digital's olfactory dispersal device (see Figure 5) along with five custom scents from Scentronix, the experience further captivates the audience's senses.

4.2 Installation

To enhance the experiential installation, we created a structure comprised of mycelial bricks. The installation (see Figure 6) was produced through a collaboration with Accenture Song Sustainability Studio and Ecovative, a company specializing in the production of mushroom mycelium bricks. The design, crafted by The Living and constructed by UrbanEarth, utilized eco-friendly, compostable materials. By introducing this physical installation alongside the experience inside of the headset, our aim is to manifest ecological consciousness to users in a tactile form and to foster a deeper appreciation for nature even before they get into the headset.

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